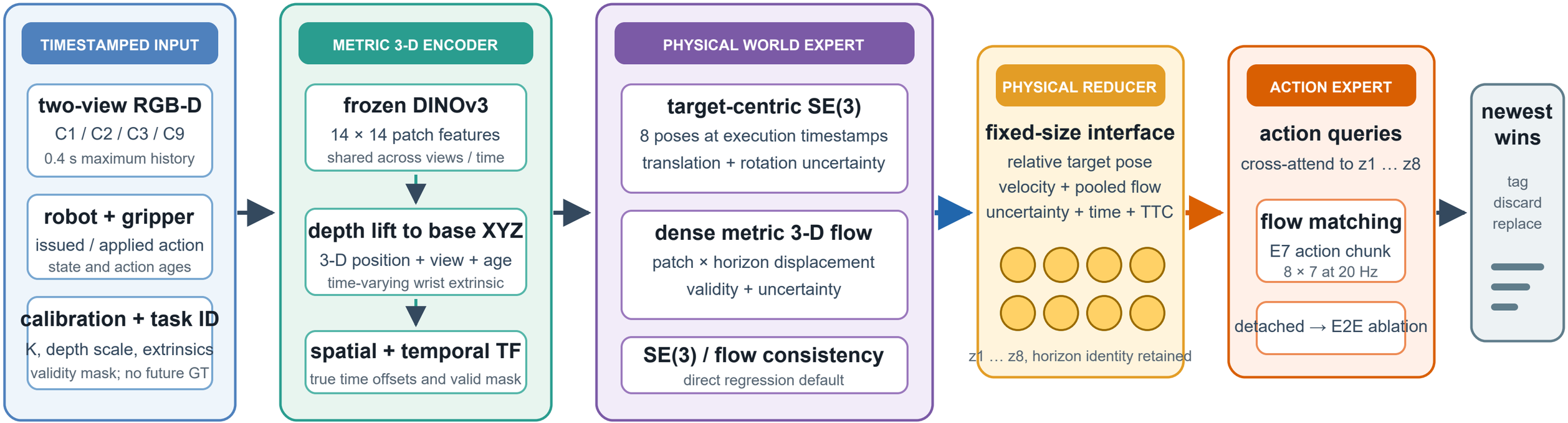
Inference: one feed-forward world pass, one-way physical bridge, continuous action execution



Training-only supervision and causal audit

metric losses

pose NLL + flow NLL + rigid consistency + action FM

future labels masked after unmatched robot contact

world decoder ablations

direct regression | standard FM | Residual-SPDF

same NFE, parameters, data, and physical horizons

intervene on z only

zero | exact-current swap | calibrated noise | oracle

measure action change, SR, safety, and timing